

Alara Martin

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Education

University of California, Berkeley, B.A. Applied Math Aug 2026 – May 2029 (Expected)

- **Relevant Coursework:** Abstract Linear Algebra, Multivariable Calculus
- **Clubs:** Education Team at Blockchain @ Berkeley, Treasurer at Cal Hiking and Outdoors Society

Foothill College, Dean's List (3.96 GPA) Sep 2024 – Jun 2026

- **Relevant Coursework:** Advanced Data Structures & Algorithms in Java, Object-Oriented Programming in C++ , Data Science, Discrete Math, Intro to Cybersecurity, Multivariable Differential Calculus

Palo Alto High School, 4.0 Unweighted / 4.48 Weighted GPA Aug 2022 – Jun 2026

- **Relevant Coursework:** Linear Algebra, Real Analysis (at Stanford ULO), AP Physics C (Mech + E&M)
- **Awards:** AP Scholar with Distinction (3x), National Debate Degree of Special Distinction, National Debate Speaking & Service Award (awarded to <1% of members), President's Volunteer Service Gold (180+ hours)

Skills

Languages & Libraries: Python, Java, C++ , JavaScript, TypeScript, HTML/CSS, LaTeX, PyTorch, React, Next.js, Pandas

Tools & Concepts: Git, REST APIs, AWS, Firebase, Vercel, Tailwind CSS, Claude Code, UI Design

Experience

AI Research Intern Jan 2025 – Present
Stanford University AI Development and Evaluation Lab *Stanford, CA*

- Achieved 94% accuracy for medical imaging QA by building and training an object detection model in PyTorch.
- Implemented Faster R-CNN keypoint detection model with ResNet-50 backbone (median error of 1.7mm per point).
- Engineered automated anonymization pipeline in Python for 1,700+ DICOM images, parsing hierarchical metadata tags to identify and scrub sensitive patient fields.
- Co-authored research paper (under review) releasing the dataset and model for automated quality assessment in medical imaging for developmental hip conditions.

Project Review Lead Sep 2025 – Mar 2026
Hack Club Athena Initiative *Palo Alto, CA*

- Reviewed 100+ technical projects; mentored participants in Git and coding best practices; developed technical tutorials and built [tutorial website](#) from scratch in vanilla HTML/CSS/JS used by 1,500+ participants.
- Ranked 24/1,500+ on the Athena coding leaderboard and selected as one of 200 hackathon invitees.

Club President & Outreach Founder Sep 2023 – Jun 2026
Girls Who Code *Palo Alto, CA*

- Developed project-based curriculum in web development and Python for 10+ peers; led a club robotics project.
- Founded student-run mentoring initiative funded by the City of Palo Alto teaching 15 elementary schoolers how to code through a 6-week program, with 100% of students completing final projects to showcase to parents.

Mathematics Tutor Jan 2024 – Aug 2026
DreamCatchers (now Pathwise) & Operation Math *Palo Alto, CA*

- Taught personalized curriculum based on diagnostic assessments of student skill gaps, ranging from advanced algebra to geometry; improved test scores by over 40%.

Projects

PKL Viewer: Python, TypeScript, VS Code API - github.com/alaramartin/pkl-viewer Aug 2025

- Developed a 24,000-user, open-source VS Code extension in Python and TypeScript enabling in-editor inspection of Python pickle files, adapting based on user feedback and collaborations through GitHub Issues/PRs.
- Designed a security-first, default disassembly mode using the pickletools library (near-instantaneous; protects against malicious payloads by avoiding code execution) and an optional load for human-readable output, weighing safety and speed with readability.

DICOM Viewer & Editor: TypeScript, VS Code API - github.com/alaramartin/dicom-viewer Aug 2025

- Built a VS Code extension with over 4,000 users across the VS Code Marketplace and Open VSX, extracting and rendering DICOM (.dcm) metadata tag trees directly in the editor view.
- Utilized native JS DICOM libraries to support in-place metadata editing, eliminating the need for external tools and enabling zero-setup installation.